9 **(a)** Fig. 9.1 shows the first three energy levels of copper.

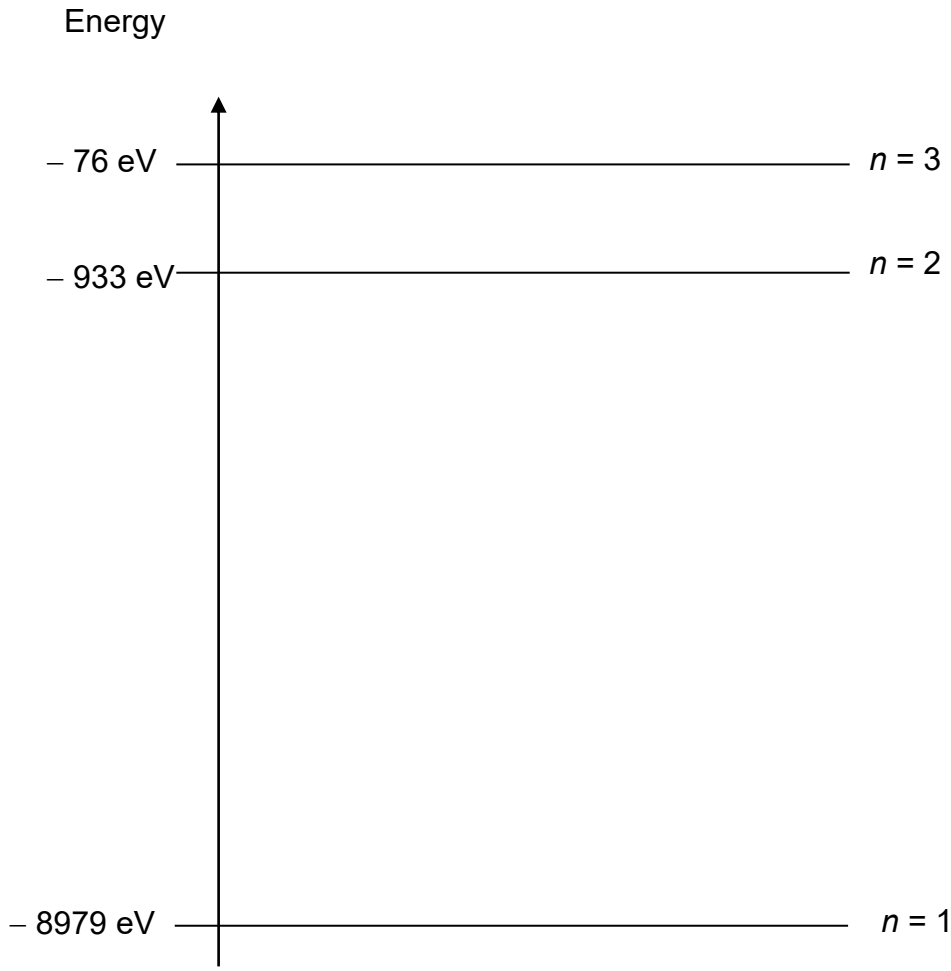

Fig. 9.1

In a particular experimental X-ray tube, a copper target is used. Electrons in the tube are accelerated by a potential difference of 10 kV before entering the target. X-rays then emerge from the target.

(i) Determine

1. the minimum wavelength of the X-ray photons.

minimum wavelength = …….. m [3]

2. the wavelength corresponding to the K-alpha characteristic line.

wavelength = ... m [2]

(ii) Sketch the X-ray spectrum of copper on Fig. 9.2 below. Use the answers in **(a)(i)** to label two important values of wavelength on your spectrum.

Fig. 9.2 [3]

(b) The diameter of a nucleus is of the order of magnitude of 10^{-15} m. Show, using the uncertainty principle, that an electron does not exist inside the nucleus.

(c) The decay of a calcium-45 nucleus (chemical symbol: Ca; Z = 20) releases a scandium-45 (chemical symbol: Sc; Z = 21) nucleus. Write a nuclear equation to deduce the type of decay that takes place.

type of decay = ……. [2]

(d) Radon-219 is a naturally radioactive element, a member of the noble gas family found in soil. Data for the α-decay of Radon-219 to form Polonium-215 are given in Fig. 9.3 below.

Nucleus	*Mass of nucleus / u*
Radon-219	219.009523
Polonium-215	214.999469
Helium-4	4.002603

Fig. 9.3

Determine the energy released during the decay.

energy released = …….. MeV [3]

(e) Bismuth-210 undergoes β-decay. The emitted β-particles have a range of energies up to a

maximum of 1.17 MeV.

Using conservation laws, explain why this range of energies leads to the suggestion that, apart from the β-particle, another particle must be emitted by the Bismuth-210 nucleus.

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